

■ Python Practice Problems

Level: **EASY**

35

Problems

Topics covered: **Lists • Tuples • Dictionaries • Functions • Loops • Conditions**

Practice • Learn • Master Python

Table of Contents

- Problem 1.** Sum of a List — List | Loop
- Problem 2.** Maximum Element — List | Condition
- Problem 3.** Count Even Numbers — List | Loop | Condition
- Problem 4.** Reverse a List — List
- Problem 5.** Remove Duplicates — List | Dict
- Problem 6.** FizzBuzz — Loop | Condition
- Problem 7.** Tuple Swap — Tuple
- Problem 8.** Word Count Dictionary — Dict | Loop
- Problem 9.** Check Palindrome — String | Condition
- Problem 10.** Factorial via Loop — Loop | Function
- Problem 11.** List Average — List | Function
- Problem 12.** Merge Two Dictionaries — Dict
- Problem 13.** Second Largest — List | Loop
- Problem 14.** Count Vowels — String | Loop | Condition
- Problem 15.** Flatten One Level — List | Loop
- Problem 16.** Square Each Element — List | Loop | Function
- Problem 17.** Tuple Element Access — Tuple
- Problem 18.** Dictionary Keys Sorted — Dict | Function
- Problem 19.** Check Prime — Loop | Condition | Function
- Problem 20.** List Intersection — List | Loop
- Problem 21.** Temperature Converter — Function | Condition
- Problem 22.** Zip Two Lists — List | Loop
- Problem 23.** Cumulative Sum — List | Loop
- Problem 24.** Count Occurrences — List | Dict | Function
- Problem 25.** Odd or Even Labels — List | Loop | Condition
- Problem 26.** Named Tuple Access — Tuple | Function
- Problem 27.** Remove Falsy Values — List | Condition
- Problem 28.** Fibonacci Sequence — Loop | Function
- Problem 29.** Default Dict Values — Dict | Function
- Problem 30.** Rotate List — List | Function
- Problem 31.** Invert a Dictionary — Dict | Loop
- Problem 32.** Multiplication Table — Loop | Function
- Problem 33.** Find Missing Number — List | Loop
- Problem 34.** String to List of Chars — String | List
- Problem 35.** Power Function — Loop | Function

Problem 1

Sum of a List

Tags: List | Loop

Given a list of integers **nums**, return the sum of all elements.

Example 1:

Input: `nums = [1, 2, 3, 4, 5]`

Output: 15

Explanation: $1+2+3+4+5 = 15$

Example 2:

Input: `nums = [10, -3, 7]`

Output: 14

Explanation: $10+(-3)+7 = 14$

Example 3:

Input: `nums = []`

Output: 0

Explanation: Empty list sums to 0

Problem 2

Maximum Element

Tags: List | Condition

Given a list of integers **nums**, return the largest element without using the built-in **max()** function.

Example 1:

Input: `nums = [3, 1, 4, 1, 5, 9]`

Output: 9

Example 2:

Input: `nums = [-5, -1, -3]`

Output: -1

Example 3:

Input: `nums = [42]`

Output: 42

Problem 3

Count Even Numbers

Tags: List | Loop | Condition

Given a list of integers **nums**, return the count of even numbers.

Example 1:

Input: `nums = [1, 2, 3, 4, 6]`

Output: `3`

Example 2:

Input: `nums = [1, 3, 5]`

Output: `0`

Example 3:

Input: `nums = [2, 4, 6, 8]`

Output: `4`

Problem 4

Reverse a List

Tags: List

Given a list **nums**, return the list reversed without using the built-in **reverse()** or slicing **[::-1]**.

Example 1:

Input: `nums = [1, 2, 3, 4, 5]`

Output: `[5, 4, 3, 2, 1]`

Example 2:

Input: `nums = ['a', 'b', 'c']`

Output: `['c', 'b', 'a']`

Example 3:

Input: `nums = [7]`

Output: `[7]`

Problem 5

Remove Duplicates

Tags: List | Dict

Given a list **nums**, return a new list with duplicates removed while preserving the original order.

Example 1:

```
Input: nums = [1, 2, 2, 3, 1]
```

```
Output: [1, 2, 3]
```

Example 2:

```
Input: nums = [5, 5, 5]
```

```
Output: [5]
```

Example 3:

```
Input: nums = [1, 2, 3]
```

```
Output: [1, 2, 3]
```

Problem 6

FizzBuzz

Tags: Loop | Condition

Given an integer **n**, return a list of strings from 1 to n where multiples of 3 are 'Fizz', multiples of 5 are 'Buzz', and multiples of both are 'FizzBuzz'.

Example 1:

```
Input: n = 5
```

```
Output: ['1', '2', 'Fizz', '4', 'Buzz']
```

Example 2:

```
Input: n = 15
```

```
Output: ['1', '2', 'Fizz', '4', 'Buzz', 'Fizz', '7', '8', 'Fizz', 'Buzz', '11', 'Fizz', '13', '14', 'FizzBuzz']
```

Example 3:

```
Input: n = 1
```

```
Output: ['1']
```

Problem 7

Tuple Swap

Tags: Tuple

Given a tuple **t** of exactly two elements, return a new tuple with the elements swapped.

Example 1:

```
Input: t = (1, 2)
```

```
Output: (2, 1)
```

Example 2:

```
Input: t = ('a', 'b')
```

```
Output: ('b', 'a')
```

Example 3:

```
Input: t = (10, 20)
```

```
Output: (20, 10)
```

Problem 8

Word Count Dictionary

Tags: Dict | Loop

Given a string **s**, return a dictionary mapping each word to the number of times it appears.

Example 1:

```
Input: s = "hello world hello"
```

```
Output: {'hello': 2, 'world': 1}
```

Example 2:

```
Input: s = "cat cat cat"
```

```
Output: {'cat': 3}
```

Example 3:

```
Input: s = "one two three"
```

```
Output: {'one': 1, 'two': 1, 'three': 1}
```

Problem 9

Check Palindrome

Tags: String | Condition

Given a string **s**, return **True** if it is a palindrome (reads the same forwards and backwards), else **False**.

Example 1:

Input: s = "racecar"

Output: True

Example 2:

Input: s = "hello"

Output: False

Example 3:

Input: s = "madam"

Output: True

Problem 10

Factorial via Loop

Tags: Loop | Function

Write a function **factorial(n)** that returns $n!$ using a loop (not recursion). Assume $n \geq 0$.

Example 1:

Input: n = 5

Output: 120

*Explanation: $5*4*3*2*1 = 120$*

Example 2:

Input: n = 0

Output: 1

Explanation: $0! = 1$ by definition

Example 3:

Input: n = 3

Output: 6

*Explanation: $3*2*1 = 6$*

Problem 11

List Average

Tags: List | Function

Given a non-empty list of numbers **nums**, return the average (mean) as a float.

Example 1:

```
Input: nums = [1, 2, 3, 4, 5]
```

```
Output: 3.0
```

Example 2:

```
Input: nums = [10, 20]
```

```
Output: 15.0
```

Example 3:

```
Input: nums = [7]
```

```
Output: 7.0
```

Problem 12

Merge Two Dictionaries

Tags: Dict

Given two dictionaries **d1** and **d2**, return a new dictionary that merges both. If a key exists in both, the value from **d2** should win.

Example 1:

```
Input: d1={'a':1, 'b':2}, d2={'b':3, 'c':4}
```

```
Output: {'a':1, 'b':3, 'c':4}
```

Example 2:

```
Input: d1={'x':10}, d2={'y':20}
```

```
Output: {'x':10, 'y':20}
```

Example 3:

```
Input: d1={}, d2={'k':5}
```

```
Output: {'k':5}
```


Problem 13

Second Largest

Tags: List | Loop

Given a list of distinct integers **nums** (length ≥ 2), return the second largest element.

Example 1:

Input: `nums = [3, 1, 4, 1, 5, 9]`

Output: 5

Example 2:

Input: `nums = [10, 20]`

Output: 10

Example 3:

Input: `nums = [1, 2, 3, 4]`

Output: 3

Problem 14

Count Vowels

Tags: String | Loop | Condition

Given a string **s**, return the number of vowels (a, e, i, o, u) it contains (case-insensitive).

Example 1:

Input: `s = "Hello World"`

Output: 3

Example 2:

Input: `s = "rhythm"`

Output: 0

Example 3:

Input: `s = "AEIOU"`

Output: 5

Problem 15

Flatten One Level

Tags: List | Loop

Given a list of lists **nested**, return a single flat list containing all elements.

Example 1:

Input: nested = [[1,2],[3,4],[5]]

Output: [1,2,3,4,5]

Example 2:

Input: nested = [['a'],['b','c']]

Output: ['a','b','c']

Example 3:

Input: nested = [[],[1]]

Output: [1]

Problem 16

Square Each Element

Tags: List | Loop | Function

Given a list of integers **nums**, return a new list where every element is squared.

Example 1:

Input: nums = [1, 2, 3, 4]

Output: [1, 4, 9, 16]

Example 2:

Input: nums = [-2, -3]

Output: [4, 9]

Example 3:

Input: nums = [0]

Output: [0]

Problem 17

Tuple Element Access

Tags: Tuple

Given a tuple **t** and an index **i**, return the element at index **i**. If **i** is out of range, return **-1**.

Example 1:

Input: `t=(10,20,30), i=1`

Output: `20`

Example 2:

Input: `t=(5, ), i=5`

Output: `-1`

Example 3:

Input: `t=(1,2,3), i=-1`

Output: `3`

Problem 18

Dictionary Keys Sorted

Tags: Dict | Function

Given a dictionary **d**, return a list of its keys sorted in ascending order.

Example 1:

Input: `d = {'banana':2, 'apple':5, 'cherry':1}`

Output: `['apple', 'banana', 'cherry']`

Example 2:

Input: `d = {3: 'c', 1: 'a', 2: 'b'}`

Output: `[1, 2, 3]`

Example 3:

Input: `d = {}`

Output: `[]`

Problem 19

Check Prime

Tags: Loop | Condition | Function

Write a function **is_prime(n)** that returns **True** if **n** is a prime number, else **False**. Assume $n \geq 2$.

Example 1:

Input: `n = 7`

Output: `True`

Example 2:

Input: `n = 10`

Output: `False`

Example 3:

Input: `n = 2`

Output: `True`

Problem 20

List Intersection

Tags: List | Loop

Given two lists **a** and **b**, return a list of elements that appear in both (without duplicates).

Example 1:

Input: `a=[1, 2, 3], b=[2, 3, 4]`

Output: `[2, 3]`

Example 2:

Input: `a=[1, 2], b=[3, 4]`

Output: `[]`

Example 3:

Input: `a=[1, 1, 2], b=[1, 2, 2, 3]`

Output: `[1, 2]`

Problem 21

Temperature Converter

Tags: Function | Condition

Write a function **convert_temp(value, unit)** that converts Celsius to Fahrenheit when unit='C', and Fahrenheit to Celsius when unit='F'. Formula: $F = C \times 9/5 + 32$.

Example 1:

Input: value=100, unit='C'

Output: 212.0

Example 2:

Input: value=32, unit='F'

Output: 0.0

Example 3:

Input: value=0, unit='C'

Output: 32.0

Problem 22

Zip Two Lists

Tags: List | Loop

Given two lists **keys** and **values** of the same length, return a dictionary mapping each key to its corresponding value.

Example 1:

Input: keys=['a', 'b'], values=[1, 2]

Output: {'a': 1, 'b': 2}

Example 2:

Input: keys=[1, 2, 3], values=['x', 'y', 'z']

Output: {1: 'x', 2: 'y', 3: 'z'}

Example 3:

Input: keys=[], values=[]

Output: {}

Problem 23

Cumulative Sum

Tags: List | Loop

Given a list of numbers **nums**, return a new list where each element is the cumulative sum up to that index.

Example 1:

```
Input: nums = [1, 2, 3, 4]
```

```
Output: [1, 3, 6, 10]
```

Example 2:

```
Input: nums = [5, -5, 5]
```

```
Output: [5, 0, 5]
```

Example 3:

```
Input: nums = [10]
```

```
Output: [10]
```

Problem 24

Count Occurrences

Tags: List | Dict | Function

Given a list **nums** and a value **target**, return how many times **target** appears in **nums** without using **list.count()**.

Example 1:

```
Input: nums=[1,2,2,3,2], target=2
```

```
Output: 3
```

Example 2:

```
Input: nums=[1,2,3], target=5
```

```
Output: 0
```

Example 3:

```
Input: nums=[7,7,7], target=7
```

```
Output: 3
```

Problem 25

Odd or Even Labels

Tags: List | Loop | Condition

Given a list of integers **nums**, return a list of strings 'odd' or 'even' corresponding to each element.

Example 1:

```
Input: nums = [1, 2, 3, 4]
```

```
Output: ['odd', 'even', 'odd', 'even']
```

Example 2:

```
Input: nums = [0, -1]
```

```
Output: ['even', 'odd']
```

Example 3:

```
Input: nums = []
```

```
Output: []
```

Problem 26

Named Tuple Access

Tags: Tuple | Function

Given a list of tuples **students** where each tuple is (name, grade), return the name of the student with the highest grade.

Example 1:

```
Input: students=[('Ali',90),('Sara',95),('Tom',80)]
```

```
Output: 'Sara'
```

Example 2:

```
Input: students=[('Zara',100)]
```

```
Output: 'Zara'
```

Example 3:

```
Input: students=[('A',70),('B',70)]
```

```
Output: 'A'
```

Explanation: Return first if tied

Problem 27

Remove Falsy Values

Tags: List | Condition

Given a list **items**, return a new list with all falsy values (None, 0, False, "", []) removed.

Example 1:

Input: items=[1,0,None,'hello','',False]

Output: [1, 'hello']

Example 2:

Input: items=[0,0,0]

Output: []

Example 3:

Input: items=[1,2,3]

Output: [1,2,3]

Problem 28

Fibonacci Sequence

Tags: Loop | Function

Write a function **fibonacci(n)** that returns the first **n** Fibonacci numbers as a list. Starts with [0, 1, ...].

Example 1:

Input: n = 6

Output: [0,1,1,2,3,5]

Example 2:

Input: n = 1

Output: [0]

Example 3:

Input: n = 2

Output: [0,1]

Problem 29

Default Dict Values

Tags: Dict | Function

Given a list of keys and a default value, return a dictionary where each key maps to the default value.

Example 1:

```
Input: keys=['a', 'b', 'c'], default=0
```

```
Output: {'a':0, 'b':0, 'c':0}
```

Example 2:

```
Input: keys=[1,2], default='N/A'
```

```
Output: {1:'N/A', 2:'N/A'}
```

Example 3:

```
Input: keys=[], default=True
```

```
Output: {}
```

Problem 30

Rotate List

Tags: List | Function

Given a list **nums** and integer **k**, rotate the list to the right by **k** positions.

Example 1:

```
Input: nums=[1,2,3,4,5], k=2
```

```
Output: [4,5,1,2,3]
```

Example 2:

```
Input: nums=[1,2,3], k=4
```

```
Output: [3,1,2]
```

Explanation: k wraps around

Example 3:

```
Input: nums=[1], k=10
```

```
Output: [1]
```

Problem 31

Invert a Dictionary

Tags: Dict | Loop

Given a dictionary **d** with unique values, return a new dictionary where keys and values are swapped.

Example 1:

Input: `d = {'a':1, 'b':2, 'c':3}`

Output: `{1:'a', 2:'b', 3:'c'}`

Example 2:

Input: `d = {1:'x'}`

Output: `{'x':1}`

Example 3:

Input: `d = {}`

Output: `{}`

Problem 32

Multiplication Table

Tags: Loop | Function

Given an integer **n**, return a list of strings representing the multiplication table for n from 1 to 10.

Example 1:

Input: `n = 2`

Output: `['2x1=2', '2x2=4', ..., '2x10=20']`

Example 2:

Input: `n = 5`

Output: `['5x1=5', '5x2=10', ..., '5x10=50']`

Example 3:

Input: `n = 1`

Output: `['1x1=1', '1x2=2', ..., '1x10=10']`

Problem 33

Find Missing Number

Tags: List | Loop

Given a list **nums** containing n distinct numbers in range $[0, n]$, return the one missing number.

Example 1:

Input: `nums = [3, 0, 1]`

Output: `2`

Example 2:

Input: `nums = [0, 1]`

Output: `2`

Example 3:

Input: `nums = [9, 6, 4, 2, 3, 5, 7, 0, 1]`

Output: `8`

Problem 34

String to List of Chars

Tags: String | List

Given a string **s**, return a list where each element is a character of **s** (excluding spaces).

Example 1:

Input: `s = "hello"`

Output: `['h', 'e', 'l', 'l', 'o']`

Example 2:

Input: `s = "hi there"`

Output: `['h', 'i', 't', 'h', 'e', 'r', 'e']`

Example 3:

Input: `s = " "`

Output: `[]`

Problem 35

Power Function

Tags: Loop | Function

Write a function **power(base, exp)** that returns base^{exp} using a loop (do not use ****** or **pow()**). Assume $\text{exp} \geq 0$.

Example 1:

Input: base=2, exp=10

Output: 1024

Example 2:

Input: base=3, exp=0

Output: 1

Example 3:

Input: base=5, exp=3

Output: 125